Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - II

Course Code: BCA104-2 **Max Marks: 100**

Course Name: DISCRETE MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Demonstrate a working knowledge of set notation and elementary set theory and recognize the connection between set operations and logic.

CO2: Apply the different properties of injections, surjections, bijections, compositions, and inverse functions.

CO3: Understand different types of matrices and operations among matrices.

CO4: Determine when a relation is reflexive, symmetric, antisymmetric, transitive, apply the properties of equivalence relations and partial orderings, and explain the connection between equivalence relations.

CO5: Effective express the elementary concepts and results of Number theory.

Section A **3*8=24**

1)	Prove that the union of two sets A and B is a superset of each set. (OR) Explain the properties of the empty set in the context of set theory.	**3**	(CO1, L2)
2)	Define and illustrate with an example a bijection function between two sets. (OR) Describe the conditions under which a function is not surjective.	**3**	(CO2, L4)
3)	Discuss the properties of transitive relations with an example. (OR) Explain how to determine if a given relation is reflexive.	**3**	(CO4, L3)
4)	State and prove the Fundamental Theorem of Arithmetic. (OR) Explain how to find the greatest common divisor using Euclid's algorithm.	**3**	(CO5, L4)
5)	Explain the concept of matrix addition and multiplication, providing an example for each. (OR) Describe the process of finding the inverse of a 2x2 matrix.	**3**	(CO3, L4)
6)	Define and give an example of an antisymmetric relation. (OR) Describe the properties of symmetric relations.	**3**	(CO4, L2)
7)	Define injective and surjective functions with examples. (OR) Explain the concept of function composition.	**3**	(CO2, L3)
8)	Describe the properties of equivalence relations. (OR) Explain how to prove that a relation is an equivalence relation.	**3**	(CO4, L4)

Section B **8*7=56**

1)	Given sets A = {1, 2, 3} and B = {3, 4}, find the Cartesian product A x B. (OR) Define and provide an example of a power set.	**8**	(CO1, L3)

2)	Prove that if f: A -> B and g: B -> C are bijective functions, then the composition g o f is also bijective. (OR) Explain how to determine if a given function is injective.	**8**	(CO2, L4)
3) a)	Prove that the set of all even integers is countably infinite. (OR) Define and give an example of a diagonal matrix.	**8**	(CO5, L4)
	OR		
b)	Show that if A and B are disjoint sets, then the power set of their union is equal to the Cartesian product of their power sets. (OR) Explain how to find the determinant of a 3x3 matrix.	**8**	(CO3, L4)
4)	Define and provide an example of a partial ordering. (OR) Explain the concept of reflexive closure.	**8**	(CO4, L3)
5)	Prove that if a relation R on set A is reflexive and transitive, then the union of R with its inverse is an equivalence relation. (OR) Define and give an example of a symmetric matrix.	**8**	(CO4, L3)
6)	Explain the concept of congruence modulo n and provide an example. (OR) Describe the process of finding the least common multiple using prime factorization.	**8**	(CO5, L3)
7)	Discuss the properties of matrix addition and multiplication. (OR) Explain how to find the transpose of a matrix.	**8**	(CO3, L4)

Section C **10*2=20**

1)	Given two sets A and B, prove that the complement of their union is equal to the intersection of their complements. (OR) Explain how to determine if a given relation is antisymmetric.	**10**	(CO1, L4)
2)	Prove that the function f(x) = x^3 is a bijection from the set of real numbers to itself. (OR) Explain how to find the inverse of a matrix using elementary row operations.	**10**	(CO2, L4)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

AI-assisted draft - teacher review and approval required before use